

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 | Kampala, Uganda (Remote)

PROFESSIONAL SUMMARY

Senior full-stack engineer (TypeScript/Node.js, React) with 6+ years building production systems where accuracy and performance are imperative. Led the zero-downtime migration of 20+ applications from Parse/MongoDB to Firebase/GCP, owning technical decisions, data modeling, and schema design end to end while reporting directly to the CTO. Kept live stock-market prices correct across 2M+ records for a fintech platform. Experienced technical leader — led a team of 4 developers and mentored engineers through documentation and training. Now applying the same end-to-end ownership and financial-data rigor to private capital analytics.

CORE COMPETENCIES

TypeScript & Node.js

React & Frontend Engineering

Database Schema Design & Data Modeling

SQL & MongoDB

API Design (REST, gRPC)

Zero-Downtime Migrations

Technical Leadership & Mentoring

GCP & Cloud Infrastructure

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer — US Remote

- **Led the end-to-end migration of 20+ applications** from Parse/MongoDB to Firebase/GCP with zero downtime, owning architecture, data modeling, and delivery while reporting directly to the CTO
- Implemented real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub — every record reconciled while production traffic ran in parallel
- Saved the company \$5,000/month by migrating all services off AWS; migrated file storage from Amazon S3 to GCS via a Python script
- Built a 3D visualisation feature using video overlay and ffmpeg, increasing buyer conversion
- Implemented Express Shipping, generating an additional \$40 revenue per order

MTailor

Mar 2022 – Jul 2022

Backend Contractor — US Remote

- Built a migration pipeline reading from MongoDB and writing to Firestore after complex data-transformation logic — schema redesign under live-traffic constraints
- Wrote documentation for fellow engineers on the new Firebase SDKs (Node.js and Python) and trained the Ops team on dashboards and backup procedures

Mind2matter

Jan 2022 – Jul 2022

Software Engineer — US Remote

- Built backends for DeFi applications using Web3 and Node.js; delivered React UIs from Figma designs under tight timelines

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer — Singapore Remote (FinTech)

- **Kept real-time stock-market prices accurate across 2M+ Firestore records**, building query jobs against Morningstar APIs for a computation-heavy financial platform where correctness was imperative
- Extended backend APIs using Firebase Cloud Functions and Express.js; built responsive React + Tailwind pages for the PWA

CodeBits

Jan 2020 – Jul 2021

Software Engineer & Team Lead — Uganda

- **Led a team of 4 developers** building legal-tech systems for NGOs — case management for FIDA Uganda and a paralegal database for LASPNET
- Architected a microservices backend using Apache Kafka, Docker, and Kubernetes; built a USSD service communicating with a Node.js backend via gRPC
- Developed a cross-platform mobile app with React Native and Expo, with Firebase push notifications

PROJECTS

Real-Time MongoDB ↔ Firestore Sync Pipeline

Live two-way sync supporting parallel production traffic during a zero-downtime migration of 20+ apps — accuracy-critical data movement with full reconciliation.

Node.js · Google Pub/Sub · MongoDB · Firestore

Healthcare WhatsApp Chatbot

Integrated OpenAI and Google Vision for prescription OCR via the WhatsApp Business API, automating a manual intake workflow.

Node.js · OpenAI API · Google Vision

Marketplace Recommendation Engine

Collaborative + content-based filtering engine in MongoDB for service-provider matching.

Node.js · MongoDB

EDUCATION

BSc Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

SKILLS

Proficient: Node.js, TypeScript/JavaScript, React, Firebase, GCP, SQL, MongoDB, Python, Git, Linux **Intermediate:** Go, AWS, Rust, React Native

Languages: English, Kiswahili